PATHI MANIKANTA

AI Evaluation Specialist | ML Engineer | RLHF/RLAIF

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PROFESSIONAL SUMMARY

AI Evaluation Specialist and Machine Learning Engineer with hands-on experience building and deploying end-to-end ML pipelines, evaluating frontier AI systems, and contributing to RLHF/RLAIF workflows. Skilled in Python, scikit-learn, TensorFlow/Keras, NLP, computer vision, model deployment, and agentic tool-use evaluation. Earned promotion from attempter to reviewer at Scale AI (Outlier) for maintaining 95%+ grading consistency across 100+ reviews. Built and deployed evaluation and ML systems, including a live agentic-trajectory auditor and a classifier reaching 92.5% accuracy with proper data leakage detection and GroupKFold cross-validation.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, SQL, HTML, CSS, Embedded C

ML & AI Frameworks: scikit-learn, TensorFlow, Keras, pandas, NumPy, NLTK, OpenCV, SHAP, Streamlit, matplotlib, seaborn

AI Evaluation: LLM evaluation, RLHF, RLAIF, rubric design, agentic trajectory review, tool-use evaluation, API evaluation, red-teaming, prompt engineering, pytest verifiers, fact-checking, data annotation

ML Techniques: Classification, regression, model training & evaluation, feature engineering, data preprocessing, signal processing, cross-validation, GroupKFold, hyperparameter tuning, data leakage detection, model deployment, model explainability (SHAP), NLP, TF-IDF, text classification, computer vision, face detection & recognition

Tools & Platforms: Git, GitHub, MATLAB, Excel, Power BI, Arduino, AWS (EC2, S3, IAM, VPC), Google Workspace, VS Code, Jupyter, pytest

Web & Frameworks: React, Node.js, REST APIs, JSON, HTML/CSS

Spoken Languages: English, Telugu, Hindi, Tamil

PROFESSIONAL EXPERIENCE

AI Evaluation Specialist, Freelance | Scale AI (Outlier) | 2026 - Present

- Promoted from attempter to reviewer within 3 months after designing single-turn agentic evaluation tasks with consistent grading accuracy and full guideline compliance.
- Authored 50+ binary pass/fail rubrics (15-25 criteria each) and developed pytest verifiers that programmatically validate task outputs using systematic test design methodology.
- Audited 100+ agentic tool-use trajectories, identifying failure patterns in call sequencing, argument accuracy, and instruction-following to improve frontier model performance.
- Architected multimodal evaluation tasks across 10+ categories, enforcing cross-modal reasoning and multi-step tool coordination for frontier AI systems.
- Contributed to the RLHF/RLAIF feedback loop training frontier AI models, pinpointing hallucination patterns, instruction-following failures, and incorrect tool-call execution.

AI Evaluation Contributor, Freelance | Handshake AI | 2026 - Present

- Authored hard, single-answer, multi-hop search questions and reference responses designed to defeat frontier models using web search, following a strict multi-rule quality rubric and a library of authoring techniques.
- Built an automated QC checker and a public workflow walkthrough that vet submissions against the rejection rules before delivery, enforcing consistency and reducing rework.

PROJECTS

TrajLens - Agentic Trajectory Auditor | LLM Evaluation Project (Deployed) | 2026 | Live Demo

- Built and deployed an interactive agentic-trajectory auditor (Streamlit) that steps through recorded tool-use traces and grades every turn against a 7-criterion binary rubric implemented as deterministic Python verifiers, tagging failures across a documented taxonomy (hallucinated tool output, malformed arguments, wrong tool selection, redundant loops).

- Built a rubric-vs-LLM-judge meta-evaluation: the deterministic rubric matches human gold labels on every trajectory while a results-only LLM-judge is fooled on hallucinated-value and retry-loop cases; keyless, zero-API-cost demo on public synthetic data.

Customer Churn Predictor | Machine Learning Project (Deployed) | 2026 | Live Demo

- Built and deployed an end-to-end churn-prediction web app: a Gradient Boosting scikit-learn Pipeline (scaling + one-hot encoding) on the 7,043-customer IBM Telco dataset, reaching 0.843 test ROC-AUC (0.847 +/- 0.013 across 5-fold cross-validation); served live on Streamlit Community Cloud.
- Engineered per-prediction explainability with SHAP (one-hot contributions aggregated back to original features) and mapped risk scores to business actions (retain / monitor / no-action) via a real-time risk gauge.

Diabetes Detection Using Foot Pressure | Final-Year B.E. Project, Anna University | 2025

- Engineered an end-to-end ML pipeline: collected data from 6 FSR sensors via Arduino across 30+ walking sessions at 20Hz, processed signals using a Butterworth low-pass filter in MATLAB, and trained classifiers in Python.
- Identified and resolved session-level data leakage inflating accuracy from 99% to a realistic 92.5% by implementing GroupKFold cross-validation, demonstrating production-level ML debugging skills.
- Extracted 80+ features (statistical, pressure distribution, temporal) and benchmarked 4 models: Random Forest (91.2%), SVM (89.7%), Logistic Regression (87.4%), Keras Neural Network (92.5%).

Spam Email Classifier | NLP Project | 2024

- Developed a complete NLP classification pipeline: text preprocessing with NLTK (stemming, stopword removal), TF-IDF vectorisation (5000 features, unigrams + bigrams), achieving 97.8% accuracy with Linear SVM on 1000+ emails.
- Benchmarked Naive Bayes, Logistic Regression, and SVM using precision, recall, F1-score, and confusion-matrix analysis for systematic model selection.

Real-Time Face Recognition | Computer Vision Project | 2024

- Implemented a real-time face detection and recognition pipeline using OpenCV (Haar Cascades + LBPH recognizer), processing live video at 15+ FPS with a 70%+ confidence threshold on a custom dataset of 50+ images per person.
- Built a custom capture-and-label pipeline and tuned the confidence threshold to balance false accepts against false rejects across varying lighting conditions.

IoT Environmental Monitoring System | Vaayusastra Internship | 2023

- Deployed a real-time sensor data pipeline using Arduino (DHT11, BMP180) streaming temperature, humidity, and pressure readings every 2 seconds to a remote dashboard, handling noisy real-world sensor data with smoothing and validation.

EDUCATION

Bachelor of Engineering, Computer Science & Engineering (Data Science) | Jansons Institute of Technology (Anna University), Coimbatore | 2021-2025 | CGPA 8.0/10

Higher Secondary (Class XII), MPC | Sri Sai Junior College, Kanigiri, Andhra Pradesh | 2021

Secondary (Class X) | Nagarjuna Model School, Kadapa, Andhra Pradesh | 2019 | CGPA 9.8/10

CERTIFICATIONS

AWS Academy Cloud Foundations, Amazon Web Services, 2022 | Microsoft 365 Productivity Suite (Advanced), Microsoft, 2022 | Ethical Hacking & Penetration Testing, Scholiverse Educare Pvt. Ltd., 2022